



Shame in the machine: affective accountability and the ethics of AI

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Abstract

The cultural weaponization of shame surrounding the use of artificial intelligence (AI) tools like ChatGPT often redirects ethical scrutiny away from systemic concerns and toward individual users. Drawing on Sara Ahmed's affect theory, this paper argues that cultural narratives of "AI shaming" function as moral displacement that redirects scrutiny away from the environmental costs, exploitative labor practices, and corporate monopolization defining contemporary AI development. The analysis examines how shame operates across academic and professional settings to create "effort anxiety" that demands both visible human labor and accelerated productivity. Current discourse treats AI use as a personal virtue problem and obscures the carbon-intensive data centers, underpaid content moderators, and proprietary knowledge systems that enable these technologies. Instead of eliminating shame, the paper proposes redirecting it toward collective accountability for AI's systemic harms. Environmental degradation, algorithmic bias, and extractive infrastructures represent the true ethical frontier of artificial intelligence. Policy frameworks, educational interventions, and governance structures offer pathways for transforming shame from individual punishment into institutional reform. The stakes extend beyond AI itself: as emerging technologies reshape society, the patterns of moral responsibility established now will determine whether innovation serves collective flourishing or perpetuates existing inequalities. Shame can become a vehicle for institutional critique and systemic accountability if we redirect its focus from individual users to the powerful corporations, governance structures, and infrastructural systems that profit from AI's rapid expansion.

Keywords Shame · Artificial intelligence · Environmental ethics · Systemic accountability · AI Ethics · Affect theory

1 Introduction

Recent advances in artificial intelligence have granted ordinary users' free access to large language models that compose persuasive essays, polished summaries, and functional code in seconds, a capability that relocates much of the labor of writing outside the human author and unsettles familiar ideas of intellectual ownership and creative effort. Unlike earlier natural language tools that performed one task per model, contemporary LLMs present a single interface that pivots from summarizing articles to writing policy briefs to explaining differential equations, thereby collapsing the distinction between drafting, revision, and reference work that once signaled human effort. Their public release through chat applications introduced instant, low-cost access to capabilities that previously required specialized training,

upending academic assessment, professional writing, and creative industries by making plausible authorship available at the click of a prompt. Philosophers and applied ethicists now grapple with questions about whether the outputs of these systems count as original thought, how statistical training on historical data reproduces bias, what kinds of invisible human labor and material extraction sustain seemingly frictionless interfaces, and how the sudden abundance of machine-generated text should be weighed against its carbon footprint (Bender et al. 2021; Crawford 2021; Benjamin 2019). Scholarly conversation condenses around three influential positions: an alignment program that seeks technical safeguards able to keep models truthful and harmless (Weidinger et al. 2022), a justice-oriented agenda that foregrounds structural inequalities and calls for participatory design with reparative policy (Crawford 2021; Benjamin 2019), and a precautionary stance that urges moratoria or strict limits until social and ecological risks are better understood (Bender et al. 2021). Each position relies on shame to police norms, although the affect itself often escapes explicit

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scrutiny: alignment research treats the user who circumvents safety rails as blameworthy, justice scholarship exposes corporations to public embarrassment for exploitative or discriminatory practices, and precautionary writing cultivates collective guilt over planetary and social costs. Attention to shame, therefore, promises a fresh diagnostic lens, because affective cues can redirect concern from systemic incentives toward personal fault, insulating the institutions that profit from large language models while intensifying moral pressure on individuals. This article interrogates the circuits through which shame circulates in contemporary debates on alignment, justice, and precaution, and it asks whether reorienting that affect from individual culpability toward collective implication can support a more effective ethics of artificial intelligence.

It is crucial to note that directing moral concern at individual AI users can mislead us away from more important issues. For decades, philosophical and cultural critiques of shame have highlighted that it often frames the self as the main site of moral failure, thereby neglecting structural factors (Ahmed 2004; Nussbaum 2004). Similarly, preoccupations with whether one is “cheating” through AI-driven programs or failing to cultivate personal virtue can overshadow pressing matters such as the environmental impact of data centers (Coeckelbergh 2020), the replication of discriminatory biases by algorithms (Noble 2018), and the increasing privatization of knowledge under proprietary machine-learning models. When the conversation is limited to personal shortcomings, more urgent collective problems go unaddressed. Instead of scrutinizing individuals for supposedly improper AI use, we might reevaluate the infrastructural, political, and economic forces shaping the conditions in which AI is designed and deployed. This redirection could shift our discourse beyond narrow criticisms of personal responsibility and toward a genuine engagement with the ways AI technology is transforming and reinforcing existing societal hierarchies.

In this paper, I argue that shame-based frameworks around AI are misguided when they focus on personal failings. Rather than centering on whether people are “cheating” or neglecting character development, we should concentrate on the shared and systemic aspects of AI: the environmental burdens created by large-scale computing, the social inequities perpetuated by data-driven tools, and the commercial monopolies made possible by proprietary AI models. While moral theories like virtue ethics offer valuable insights into individual character, they cannot alone address or overturn the structural factors that prompt reliance on AI in the first place. Shifting to a collective perspective that recognizes our interconnected dependence on technology opens pathways for more robust accountability mechanisms. Such a recalibration not only prompts us to think about the ethical implications of AI use at a societal level, but also provides a

framework for holding powerful institutions, corporations, and governing bodies responsible for the ways they develop and deploy these technologies. By redirecting shame from individual users to the systems that prioritize profit or convenience over equity and sustainability, we can begin to tackle the deeper challenges that define our AI-driven world.

2 Artificial shame

In the context of AI use, shaming often appears as an accusation that the user is avoiding genuine effort and failing to live up to ideals of authentic and creative thought. Such attitudes align with what Giray (2024) terms ‘AI shaming,’ a cultural phenomenon where the use of AI tools is criticized as deceitful, lazy, or inherently inferior to human effort. AI shaming in nine recurring accusations levelled at AI users: AI-mediated work lacks authenticity, carries moral taint, cheapens expertise, threatens employment, and promotes superficial engagement with disciplinary standards (Giray 2024). This practice often manifests across academic and professional settings as discouraging open dialogue about responsible AI integration and fostering an environment of secrecy and anxiety. AI shaming nudges individuals to hide or deny their use of AI for fear of being disqualified from the professional or academic norms that dictate authenticity and rigor. At the same time, the shame linked to AI does more than stigmatize a single choice; it also sidesteps the larger forces driving AI adoption by placing the onus on individual users. Thus, the cultural tendency to view AI reliance as a personal failure diverts attention from the broader systems that condition or even encourage AI integration.

AI shaming is especially visible in academia, with worries about plagiarism and “cheating” blending with longstanding anxieties over technological shortcuts. Critics frequently warn that students might rely on AI-generated text for coursework and thereby bypass meaningful learning. Many educators worry that large language models erode the epistemic and ethical foundations of assessment by supplying students with ready-made prose that can be submitted as original work (Sullivan et al. 2023). A survey of one hundred news reports published in English-speaking countries within the first months of ChatGPT’s release observed that articles about it concentrated overwhelmingly on allegations of cheating, institutional bans, and fears about the demise of traditional essays (Sullivan et al. 2023). This study argued that such alarm frequently eclipses discussion of the technology’s potential to democratize learning through scaffolding, plain-language explanations, or assistance for multilingual writers. The seemingly unending warnings about AI-generated coursework functions as a shorthand for deeper concerns about authenticity and merit that undermine the legitimacy of existing grading systems. Citation density,

syntactic complexity, or discipline-specific jargon, and other traditional markers of effort may still appear in the text, but they no longer reliably indicate real engagement or genuine understanding making it harder to justify grading systems that evaluate only the final product rather than the process behind it. Additionally, artificial intelligence amplifies long-standing patterns of academic misconduct by lowering the time and skill required to fabricate plausible submissions (Birks and Clare 2023). In short, the critics fear not only plagiarism in the narrow sense but a broader displacement of intellectual development by turnkey composition tools that conceal ignorance beneath grammatical polish.

Shaming dynamics extend beyond classrooms to professional settings, with AI use impacting reputations that can determine career trajectories. Experimental research that presented thousands of participants with hypothetical co-workers reports a consistent “social evaluation penalty” for employees who disclose habitual chatbot use, with respondents rating those employees as less competent, less diligent, and less trustworthy even when work quality remains identical (Reif et al. 2025). Negative talk like allegations, slurs, and quality-based blaming about AI is not mainly about technical or artistic defects but instead serves as a boundary that safeguards the cultural and material privilege of knowledge workers (Sarkar 2025). Creative fields also amplify the moral charge by treating AI assistance as a direct affront to authenticity. One analysis of text-to-image generators argued that every prompt appropriates aesthetic labor and, therefore, marks the user as complicit in exploitation, a claim that recasts artistic experimentation as an ethical lapse (Goetze 2024). Earlier scholarship on language models warns of “ersatz fluency” and positions writers who incorporate generated prose as negligent stewards of cultural discourse, once again linking the presence of AI to personal irresponsibility rather than to questions about craft standards (Bender et al. 2021). Together with institutional rules and detection tools, these discursive practices portray AI-assisted output as uncreative, harmful, or unethical and police who may enter professional domains and, SUCH AS historic sumptuary laws and protectionist policies, the real motive is to preserve an exclusive class identity rather than to address genuine harms. This policing imposes psychological burdens, amplifies racial bias, and chills the participation of underprivileged groups (Sarkar 2025). Such judgments of AI use do not question the effectiveness of the tool; they simply attach a fault of character to anyone who admits to using it.

While the cited concerns regarding the ethics of AI use have merit, they often focus narrowly on personal integrity and imply that students must simply resist temptation through greater discipline. By doing so, they overlook deeper systemic issues that make AI tools attractive, such as the corporate interests profiting from the data that feeds AI models or the institutional priorities that favor rapid

productivity over genuine intellectual engagement. As a result, moral panic and shame take center stage and place blame on users rather than on the frameworks that normalize and sometimes incentivize these alleged “shortcuts.” This narrow approach not only fails to address structural concerns but also exacerbates the stigma surrounding AI and makes open dialogue about responsible usage even more difficult.

3 Sticking keys

Shame is usually depicted as an inward surge of embarrassment that flares up when a person imagines being judged for a flaw or wrongdoing, a feeling that belongs to private mental life and fades once the moment passes. In *The Cultural Politics of Emotion*, Sara Ahmed theorizes shame beyond its solitary sensation and instead according to its effect of rearranging relations among bodies, norms, and communities; her account treats shame as an orientation that directs attention and movement toward the standards that seem to have been violated while simultaneously displaying the gap between the self and those standards (Ahmed 2004). Consider an ordinary scene in which a child knocks over a glass at dinner and feels cheeks heat, shoulders curl, and eyes drop toward the table; the body not only registers failure but also bends toward the parental norm of carefulness that now feels out of reach. The same dynamic appears when a student fears plagiarism accusations or when a worker worries that reliance on an automated tool signals laziness. Shame exposes deviation, but the affect is both punitive and potentially transformative as it promises reintegration if the subject redoubles effort to meet the ideal just revealed making. Each attempt to repair the breach carries the risk of renewed exposure, so shame binds individuals to a repeating cycle of self-scrutiny in pursuit of belonging and always before an actual or imagined audience whose recognition is required for legitimacy. By foregrounding this relational and temporal structure, Ahmed equips us to analyze how ideals of effort, authorship, and merit become affectively enforced in debates about artificial intelligence.

Ideas about what makes someone a “good” student shape how people are judged before they even start their work. Some students are given the benefit of the doubt and seen as competent from the beginning, while others must prove themselves by showing visible and often exhausting effort just to be taken seriously. Instead of just responding to the work itself, these expectations shape who is seen as credible in the first place. Normative ideals, like those defining “good” students, gain authority through shame (Ahmed 2004). One feels shame from senses one’s own insufficiency in the face of the normative ideal of tireless productivity and, in doing so, makes that ideal even more authoritative. These normative ideals of productivity are not experienced

equally. Black life is repeatedly summoned to perform “wake work” as a ceaseless practice of evidencing humanity for an audience that doubts it, so effort itself becomes a credential demanded precisely from those whom society places at the margins (Sharpe 2016). The devotion to visible productivity illustrates what Lauren Berlant calls “cruel optimism”—an attachment to ideals that promise security and belonging while simultaneously blocking the very flourishing they advertise (Berlant 2011). The attachment turns cruel when goals like academic success or professional recognition demand practices of self-regulation that exhaust the subject and make promised relief unreachable. Generative artificial intelligence intensifies this bind by appearing to offer effortless productivity while threatening to undercut the moral weight once carried by labor itself. Shame thrives in this atmosphere by policing any reliance on AI as a failure to live up to the dream that diligent effort will secure stable value, illustrating how attachments persist not because they deliver on their promise but because abandoning them would require facing a landscape where worth can no longer be proved by hard work alone.

A parallel mechanism in the academy presses women of color to translate experiential knowledge into dense theoretical vocabulary to satisfy institutional gatekeepers and thereby turn intellectual labor into an ordeal of proof rather than an exploration of ideas (Christian 1988). Citations themselves can become a public test of belonging as well insofar as scholars whose work challenges dominant frames are shamed as parochial or insufficiently rigorous unless they adopt the stylistic markers of the canon (Ahmed 2012). Across these examples, shame instructs subjects to overperform diligence to reveal process as spectacle and to align outputs with authorized forms and transform effort itself into a moral drama that allocates worth. The same structure reappears in contemporary conversations about artificial intelligence when suspicion of machine assistance intensifies the demand for demonstrable human toil so those marked without genius remain burdened by the obligation to prove every minute of their work.

Generative text systems such as ChatGPT that let a writer move from prompt to finished prose in seconds by removing the visible stages of outlining, drafting, and revision that teachers long treated as proof of honest work and steady effort (Cotton et al. 2023). Without that trail, faculty report treating any polished submission as suspicious unless the student can produce prompt logs or earlier drafts creating a pattern documented across university academic integrity officers (Lukac and Lazareva 2023). Universities now circulate policies that require students to paste full chat transcripts into appendices or to draft assignments in monitored, in-class settings, and detection services advertise algorithmic checks that flag writing with no observable process as presumed misconduct (Turnitin 2023). Reviews of these

measures conclude that the core worry is not technical accuracy but the sudden collapse of effort as a moral signifier: when software can supply fluent text, legitimacy shifts from the quality of the argument to the visibility of labor, so anxiety concentrates on who used AI and how rather than on what the text actually accomplishes (Kofinas et al 2025). The contradiction between celebrating AI productivity while demanding visible human effort produces what might be called “effort anxiety,” a fear of being caught either doing too little by letting the model think or doing too much by refusing the tool and falling behind escalating benchmarks. Faculty describe students who submit flawless essays accompanied by lengthy disclaimers that catalogue every prompt in an attempt to prove diligence despite the efficiency of the result. Meanwhile, project managers report workers who rely on generative code yet delete the tool’s suggestions from commit histories to claim personal craftsmanship. Abstaining does not resolve the tension; professionals who refuse machine assistance often narrate their abstinence as a badge of integrity, but performance metrics still penalize them when productivity lags behind automated peers and turn righteousness into a practical disadvantage. The moral economy of artificial intelligence, therefore, rests on ambivalence: reliance induces guilt because effort remains a public virtue, and refusal induces anxiety because acceleration has become a private survival strategy.

Shame operates by orienting the subject toward an ideal, exposing the gap between lived reality and that ideal, and narrowing perception until the self seems solely responsible for closing the gap (Ahmed 2004). Shame’s inward turn gives secrecy its appeal. Once an individual reads their reliance on artificial intelligence as a personal shortfall, concealment appears prudent because it promises protection from judgment while preserving the fantasy of autonomous effort. Secrecy, therefore, manifests the isolating logic of shame rather than originating it, and any ethical analysis that begins with the hidden use of language models must recognize that it is responding to a symptom rather than to the root affective structure. The movement from exposure to concealment is predictable due to the demands of public life. According to Michael Warner, public discourse calls speakers to appear before strangers as rational and self-governing agents stripped of any markers of vulnerability or dependence, and a participant can only claim the credibility attached to the public voice by passing through this purification (Warner 2002). Everything that signals situated need, from fatigue to relying on help, must be relocated to a private sphere where it can be managed out of view. Shame attaches to that private side and polices the boundary and warns that any leak of the hidden material will compromise the speaker’s legitimacy. Because the division is effective as well as spatial, the act of hiding not only follows shame but reproduces it by confirming that the concealed practice

is indeed unfit for public display. Warner, therefore, shows that secrecy and shame reinforce each other; the pull toward privacy protects dignity in the short term while entrenching a system that marks dependence as disqualifying. Generative artificial intelligence moves naturally into this public–private machinery by finding relief in secrecy that simultaneously leaves institutional workloads and assessment norms unchallenged. The concealment verifies the reigning story that legitimate labor is fully individual, so the structural pressures that make machine help attractive remain invisible. Secrecy, in other words, is not the cause of shame around AI but its most visible footprint and addressing it requires shifting focus from individual concealment to the collective arrangements that assign blame to the private self while allowing public ideals of autonomous productivity to stand untouched.

These dynamics of concealment and exposure extend across creative and professional fields wherever intellectual labor is valued. Writers who lean on large language models to draft publicity copy or grant proposals append process notes or partial transcripts to restore the moral link between labor and text (Kofinas et al. 2025). Independent musicians who experiment with AI stem separation or automated mastering face online comments suggesting that “real” producers still mix by ear, prompting many to post behind-the-scenes videos documenting every manual adjustment to prove craft competence. Software developers using code-completion tools like GitHub Copilot often strip automated suggestions from commit messages before pushing to public repositories, signaling human expertise even as repository analytics reveal rising dependence on machine-generated snippets. Across these fields, shame sticks when assistance remains hidden, and visibility of process becomes a token of legitimacy that follows existing hierarchies of status and trust.

4 Restarting the system

Reorienting shame around AI use calls for an expanded perspective that goes beyond labeling individuals as wrongdoers. Too often, popular narratives fixate on whether a student “cheated” by employing ChatGPT or whether a content creator “betrayed” their creative calling, while ignoring the environmental, economic, and sociopolitical structural forces that shape how AI is designed and deployed. When recast as a lens that reveals these systemic issues, shame has the potential to shift attention away from personal guilt and toward collective accountability. For instance, directing shame at large-scale problems, such as carbon emissions, data monopolies, and algorithmic bias, could spur institutional reforms that require sustainable infrastructure, equitable data governance, and fair algorithmic standards (Strubell et al. 2019). Rather than fostering secrecy or stigma, this

orientation treats shame as a communal impetus for ethical AI practices, steering the conversation toward how best to reconcile technological progress with social and ecological responsibilities.

The environmental consequences of large-scale AI systems constitute a critical yet frequently overlooked dimension of AI ethics. Unlike discussions that fixate on whether a user has “cheated” by employing ChatGPT, the ecological toll of AI underscores how moral responsibility extends far beyond personal behaviors. Data centers, which power everything from mundane file storage to the most advanced neural networks, serve as one of the most glaring examples of how AI’s infrastructure exacts a collective toll. According to recent estimates, data centers currently account for approximately 1–1.5% of global electricity consumption, with some analyses warning that this figure could escalate if AI adoption continues to expand unabated (Crawford 2021; Schwartz et al. 2020). A 2023 report from the International Energy Agency (IEA) projects that the electricity demand from global data centers, cryptocurrency mining, and artificial intelligence could double by 2026. If this trend continues, the combined usage will exceed 1,000 terawatt-hours annually, which is roughly equivalent to the total electricity consumption of Japan (IEA 2023). The training of large language models plays a significant role in this increase. For example, training GPT-3 consumed an estimated 1,287 megawatt-hours of electricity and produced more than 500 metric tons of carbon dioxide emissions (Patterson et al. 2021). Developers frequently highlight improvements in performance and model accuracy, but the environmental consequences of this training remain hidden from users. This lack of transparency allows public discourse to focus moral judgment on individuals, rather than addressing the institutional and technological systems that drive resource-intensive AI development. To reframe shame effectively, we must confront the structural conditions that reward computational scale and speed while ignoring ecological limits and planetary costs. In the United States alone, data centers may already be using more than 70 billion kilowatt-hours (kWh) of electricity each year or enough to power millions of homes (Schwartz et al. 2020). These figures manifest a broader systemic issue: as AI matures, companies build increasingly expansive data infrastructures that require continuous energy input and specialized cooling systems, thereby contributing to carbon emissions and resource depletion.

The training of large language models further amplifies this environmental burden, particularly because these models rely on massive clusters of graphics processing units (GPUs) or tensor processing units (TPUs) running for extended periods (Bender et al. 2021). Researchers regularly tweak hyperparameters and retrain models in iterative cycles, resulting in a cumulative carbon footprint that can be

staggering. For instance, one study indicated that developing and tuning a single transformer-based model could emit as much as 284 metric tons of CO₂—roughly equivalent to five times the lifetime emissions of an average automobile in the United States (Strubell et al. 2019). Moreover, generative AI initiatives commonly demand repeated experiments in pursuit of incremental performance gains, ratcheting up the ecological toll in ways often invisible to end users. This dynamic epitomizes the systemic character of AI’s ethical problems: while individuals might be shamed for deploying AI in everyday tasks, the hidden supply chains and energy-hungry infrastructures that make such deployments possible go largely unexamined.

Critically, this environmental toll intersects with economic and social inequities, with resource-intensive data centers frequently drawing on electricity grids in regions with weaker environmental regulations (Noble 2018; Mohamed et al. 2020). The result can be a form of “green colonialism,” where AI benefits are predominantly reaped by global corporations headquartered in the Global North, while emissions and strain on local infrastructures are disproportionately offloaded onto more vulnerable communities (Benjamin 2019; Crawford 2021). For example, data centers have been increasingly concentrated in areas like rural Chile and South Africa, where energy regulations are weaker and water access is more easily negotiated by foreign corporations. In Chile, Google’s data center operations in the Atacama region have sparked controversy for consuming enormous volumes of groundwater in one of the world’s most arid zones, raising concerns about environmental justice and water sovereignty. Similarly, Microsoft’s data center expansions in South Africa benefit from tax incentives and regulatory exemptions, allowing corporations to externalize ecological costs while consolidating infrastructural control in the Global North (Crawford 2021). These infrastructures draw on local power and water resources while serving global cloud platforms headquartered in the United States and Europe and exemplify a postcolonial asymmetry in AI development: extractive infrastructures and externalized costs occur largely in the Global South, while benefits accrue to the Global North. A related issue emerges in the outsourcing of data labeling and content moderation labor. Kenyan workers, for instance, were paid as little as \$1.32 per hour to review and filter traumatic content for OpenAI’s safety layer, enduring harmful psychological effects with few protections (Perrigo 2023). These examples clarify that global AI infrastructures rely not only on planetary-scale energy and data but also on human labor embedded in exploitative geopolitical hierarchies. To frame shame solely in terms of Western user behaviors is to erase the full chain of inputs and consequences that define contemporary AI ecosystems. Such a pattern reveals that environmental ethics in AI cannot be divorced from questions of justice

and collective responsibility. The prevailing narrative that shames individuals for “misusing” AI and thereby obscuring the overarching systems that enable profit-driven exploitation is part of the problem, as it detracts from mobilizing meaningful regulatory efforts that could curb emissions at scale.

In response to these challenges, various initiatives for sustainable AI development have begun to surface, though they remain nascent and unevenly applied (Coeckelbergh 2020). On the policy front, stricter reporting guidelines and carbon taxes on data centers could prompt greater transparency about the true energy costs of AI applications (Strubell et al. 2019). Additionally, emerging research on “Green AI” focuses on devising algorithms and hardware optimizations that reduce computational overhead without substantially compromising model performance (Schwartz et al. 2020). Companies, universities, and governmental agencies are also exploring renewable energy sources for powering data centers in regions where wind or solar power is abundant. Such measures, however, require sustained systemic commitment and cannot be reduced to moral admonitions aimed at single users. Instead, collective oversight of corporate practices, the integration of sustainability benchmarks into AI research funding, and global cooperation on emissions standards are imperative to mitigating the planetary costs of AI (Crawford 2021; Mohamed et al. 2020). Foregrounding the environmental impact of AI illuminates how shame can be reoriented away from individual users and toward large-scale, energy-intensive infrastructures that perpetuate ecological harm. This shift not only strengthens the argument for systemic accountability but also clarifies why moralizing AI usage as a personal defect overlooks the broader contexts of production and consumption. Addressing the carbon footprint of training large language models and the socioeconomic imbalances tied to data center siting requires concerted policy actions, corporate transparency, and innovative research—none of which can be achieved through penalizing end users alone. In this sense, spotlighting AI’s environmental consequences invigorates the call for collective ethics, bridging moral discourse and institutional responsibility in ways that transcend the narrow lens of individual shame.

One approach to this reframing involves comprehensive educational efforts that highlight AI’s hidden operational layers. Instead of singling out those who use AI for convenience, educational programs could integrate digital literacy with ecological literacy to ensure that both students and professionals understand how data centers consume resources and emit carbon. Such transparency reframes shame from a judgment of personal character to a recognition of the larger infrastructural costs linked to AI and prompt institutions to assume responsibility for technological impacts. Policy interventions could buttress these efforts by mandating clearer reporting on AI’s resource usage, imposing stricter

environmental standards on data centers, or incentivizing low-carbon machine-learning models. At the same time, communities, advocacy groups, and consumers can collectively call attention to the decisions made by tech firms and public agencies, using moral pressure and civic engagement to demand more responsible AI deployment.

5 Policy and governance

Any effort to reposition shame as a tool for collective accountability also needs to engage with the legal and policy frameworks that guide AI's development and use. While cultural narratives around AI often center on individual wrongdoing or personal virtue, genuine reform requires the support of robust governance structures. Various governments and international bodies have proposed regulations that could, in principle, address systemic inequalities and ecological burdens, although their effectiveness remains a point of debate. One familiar example is the EU AI Act, which classifies AI systems into categories of minimal, limited, high, or unacceptable risk and assigns corresponding regulatory obligations. High-risk systems must comply with transparency requirements, documentation standards, human oversight protocols, and post-deployment monitoring. Systems deemed to pose an unacceptable risk, such as real-time biometric surveillance in public spaces or social scoring applications, are prohibited outright. Although the Act marks an important development in binding AI regulation, it has drawn criticism for its vague risk definitions, limited enforcement capacity, and susceptibility to corporate lobbying during its drafting process (Crawford 2021). In contrast, the United States does not have a centralized framework for AI regulation. Instead, it relies on a fragmented set of voluntary guidelines and sector-specific rules. The 2022 White House Blueprint for an AI Bill of Rights outlines principles such as privacy, fairness, and transparency, but it does not carry the force of law and includes no mandatory compliance mechanisms (White House 2022). These contrasting approaches highlight the uneven global regulatory environment and reinforce the need for enforceable accountability measures that apply across jurisdictions and prioritize the interests of affected communities. By mandating transparency, oversight, and impact assessments, the EU AI Act attempts to shift at least part of the moral and legal onus away from individual users to target the corporate and infrastructural actors behind AI systems. Critics note, however, that even well-intentioned regulations can be undermined by vague language, corporate lobbying, and a lack of enforcement mechanisms (Crawford 2021). Other regions have taken different approaches: the U.S. has relied heavily on a patchwork of sector-specific guidelines, while China's governance model emphasizes strong state oversight

with potential implications for human rights. Internationally, the OECD AI Principles and UNESCO's Recommendation on the Ethics of Artificial Intelligence highlight the importance of fairness, transparency, and accountability, echoing many of the concerns raised in this paper. Nevertheless, these documents often lack binding force, leaving significant gaps in global AI governance.

To channel shame into more systemic reform, policy-makers could mandate ongoing transparency reporting for AI systems and require organizations to disclose energy consumption, algorithmic decision-making processes, and data provenance. These measures would expose the hidden infrastructures fueling AI and making it more difficult for corporations to externalize environmental costs or perpetuate biases unseen. Another option involves incentivizing "green AI" (Strubell et al. 2019), encouraging research into low-carbon approaches to model training and deployment, and penalizing companies whose practices exceed certain emissions thresholds. Ethical AI development could also be driven by public funding strategies. Governments can prioritize grants for open-source AI projects and require that recipients adhere to fairness, privacy, and sustainability guidelines (Noble 2018). In addition to these carrots, policy "sticks" might include penalties for noncompliance, such as meaningful fines for corporations that fail to meet established benchmarks or that conceal key information about AI systems. Integrating these measures into existing regulatory structures would complement broader efforts to redirect shame from individuals to institutional actors, fostering a shared sense of responsibility for mitigating AI's harms.

Despite these potential interventions, AI governance remains fraught with global asymmetries. Many resource-intensive AI initiatives originate in the Global North, while their negative externalities, like environmental damage or exploitative labor practices, are disproportionately borne by regions in the Global South (Mohamed et al. 2020). Crafting policies that apply across national boundaries is complicated by divergent political systems, economic priorities, and legal traditions. There is also the challenge of preventing regulatory "race to the bottom," wherein regions compete to attract AI investments by offering lax oversight. Ultimately, addressing these disparities necessitates transnational cooperation, potentially in the form of international treaties or frameworks that recognize AI's global ramifications. This might involve creating a global AI oversight body empowered to set minimum standards for environmental impact, algorithmic transparency, and labor conditions. Although the realization of such initiatives remains uncertain, they align with the broader goal of harnessing shame as a collective force by pressuring not just individual users but multinational corporations and governments to uphold more equitable and sustainable AI development practices. By embedding concrete policy measures into a discourse

of collective accountability, we can reinforce the idea that moral blame for AI's excesses need not be pinned on a single user. Instead, an integrated governance approach can reshape both the cultural narratives and the institutional frameworks around AI, making it more feasible to tackle underlying harms, ecological, social, and otherwise, on a structural level.

6 Fostering sustainable and equitable accountability

In recent years, the velocity of AI innovation has prompted an equally urgent conversation about how to shape its trajectory in a manner that promotes fairness, transparency, and sustainability. The rapid evolution of machine learning and generative models signals not just an intensification of AI's capabilities but also a transformation of the social and environmental landscapes in which these technologies operate. By considering emerging trends in AI development, envisioning potential governance scenarios, and foregrounding the critical role of education, we gain a forward-looking perspective on how best to cultivate ethical AI systems that serve a collective good. A key emerging trend revolves around the shift from specialized to increasingly "general" AI applications as a movement exemplified by ever-more capable language models, multi-modal systems (combining text, image, and speech), and ongoing research in reinforcement learning (Vallor 2016; Bender et al. 2021). While these models promise heightened adaptability and broader utility across fields such as healthcare, environmental modeling, and education, they also magnify moral and political questions about how algorithmic decisions are made. As AI grows more complex, the risk of embedding and perpetuating biases likewise increases, especially when training data replicate and reinforce historical or systemic injustices (Benjamin 2019; Noble 2018). Moreover, the potential for misinformation and disinformation campaigns escalates when content-generation becomes seamless enough to mimic human language patterns (Zuboff 2019). Hence, the future of AI ethics will hinge on ensuring transparency in model architectures and data provenance, requiring both regulators and the AI community to adopt measures such as standardized audits, adversarial testing for bias, and more conscientious data stewardship (Crawford 2021).

Against this backdrop, at least two broad scenarios for AI governance and accountability appear possible. In one scenario, market forces continue to dominate, with corporate interests shaping AI's trajectory and regulatory frameworks lagging behind. Under such conditions, commercial entities may further monopolize data resources, limiting opportunities for public scrutiny and entrenching patterns of social inequality (Mohamed et al. 2020). Mechanisms

of accountability may remain fragmented and limited to sporadic legal interventions or voluntary ethical charters that lack enforceable standards. In a second, more hopeful scenario, governments, intergovernmental bodies, and civil society groups collaborate to establish robust, enforceable regulations, including mandatory environmental reporting on AI's carbon footprint, legally binding commitments to mitigating algorithmic discrimination, and frameworks for protecting user privacy (Strubell et al. 2019; Crawford 2021). Such governance might also promote open-source innovations to democratize AI research, thus reducing the power asymmetries that currently favor a small number of dominant tech corporations (Noble 2018). A key facet of this scenario would be the active involvement of an informed public, whose demand for transparency and equity in AI systems aligns with stronger institutional oversight through a collective ethic that moves beyond shaming individuals and instead holds corporations and regulators answerable to wider social and environmental imperatives.

A crucial but often underestimated element in propelling AI toward an ethical future is the role of education. As earlier sections of this paper have demonstrated, "AI shaming" discourages many from discussing or even learning about AI's functionalities, pitfalls, and potential benefits (Giray 2024). Expanding formal and informal educational programs at multiple levels ranging from primary schools to continuing professional development could help dismantle the stigma around AI usage, while simultaneously promoting a more accurate understanding of AI's systemic impacts. At the K–12 level, educational initiatives have begun integrating digital ethics and ecological literacy into classroom instruction. For instance, the MIT Media Lab's "AI + Ethics" curriculum introduces middle and high school students to concepts such as algorithmic fairness, data bias, and the environmental costs of training large models (MIT Media Lab 2020). Stanford's "CS Ethics in Society" course similarly challenges undergraduate students to conduct carbon audits of AI systems and assess the ethics of data supply chains. Through project-based learning, students are asked to analyze labor conditions in data annotation firms and to propose policy interventions to mitigate those harms. These models demonstrate how ecological and social critique can be seamlessly embedded into STEM curricula to challenge the privatized and depoliticized image of AI. Similarly, the World Economic Forum's "AI for Youth" program has partnered with ministries of education to promote ethical AI awareness in public schools across India and Latin America (World Economic Forum 2021). These models can serve as starting points for broader adoption. A redesigned high school curriculum might include case studies on carbon emissions from AI infrastructure, lessons on data justice, and project-based assignments that evaluate AI tools from environmental and social perspectives. Rather than treating

AI as a neutral technological artifact, students would be encouraged to understand its planetary consequences and social entanglements.

At the university level, interdisciplinary programs that combine computer science, environmental studies, and philosophy could train future developers, ethicists, and regulators to think systemically. Courses might ask students to calculate the resource intensity of machine learning models, critique the labor conditions of data labeling, or assess the transparency policies of tech firms. These proposals echo Schwartz et al.'s (2020) call for “Green AI” approaches that prioritize efficiency, sustainability, and openness in computing research. Such curricular reforms would reorient education away from personal virtue signaling and toward critical engagement with the infrastructures that make AI both possible and problematic. At the tertiary and professional levels, interdisciplinary programs that blend computer science, social sciences, and philosophy would equip the next generation of AI specialists with the capacity for reflexive self-critique and holistic impact assessments (Vallor 2016). Building AI literacy in this way creates a more discerning public, one capable of demanding responsible corporate and governmental behavior and of redirecting shame toward structural issues rather than individual failings.

Looking ahead, this combination of inclusive governance, technologically informed regulation, and widespread ethical literacy forms the crux of AI's sustainable future. In centering structural accountability over personal blame, we invite a paradigm in which AI tools become catalysts for human flourishing and ecological stewardship, rather than accelerators of inequality or environmental degradation (Benjamin 2019; Crawford 2021). Yet realizing this vision depends on the collective commitment of governments, corporations, educational institutions, and civil society alike. Whether through binding legislation, transparent AI audits, or globally recognized guidelines that prioritize environmental and social justice, the pathways remain open for harnessing AI's vast potential while mitigating its risks. Only by coupling robust governance frameworks with a culture of systemic accountability can we move past the misplaced shame currently afflicting individual AI users and, instead, transform shame into a powerful motivator for large-scale reform.

7 Conclusion: touch grass

The pervasive sense of shame surrounding ChatGPT and other AI tools is not without merit; it reflects a collective unease with the rapid technological changes reshaping our work and social lives. However, as this paper has argued, such shame is routinely misdirected when it is pinned on individual users instead of the systemic forces that govern AI's development and deployment. By focusing on personal

lapses framed as laziness, cheating, or moral insufficiency, public discourse obscures the environmental toll of large-scale computational infrastructures and the broader social inequalities entrenched by proprietary, data-driven systems. More productively, shame can function as a powerful communal affect, activating moral and political energies that target the root causes of AI's most pressing problems, including unsustainable energy consumption and algorithmic biases.

Reframing shame as a collective tool rather than an individualizing condemnation highlights the potential for new forms of social and institutional accountability. Redirected shame can drive public demand for transparent reporting on AI's carbon footprint, more rigorous regulatory oversight, and the prioritization of greener, more equitable research agendas. Historical precedents show this reorientation works: public shaming of fossil fuel companies' climate impacts spurred divestment movements and carbon disclosure laws. Similarly, collective outrage over digital surveillance birthed GDPR's consent frameworks. Applying this model to AI could transform shame from personal stigma into fuel for structural reform. When supported by policy interventions and shifts in cultural narratives, these strategies promise a healthier alignment of moral concerns with the material realities of AI production. Instead of stifling technological adoption or stoking secrecy around AI use, this collective orientation has the potential to foster genuinely ethical and sustainable AI practices. By implementing these interventions, we not only address the immediate harms caused by AI systems but also lay the groundwork for a broader cultural shift in how society confronts emerging technologies. This realignment of shame toward systemic accountability and away from individual blame helps to foster the collective vigilance necessary to navigate the ethical complexities of new innovations such as quantum computing, genomics, and autonomous systems. Such vigilance ensures that the moral energies activated by shame are harnessed for structural reform rather than punitive moralism.

The implications of such a reframing extend well beyond ChatGPT or the realm of generative AI. As emerging technologies, from genomics to quantum computing, gain traction in society, we must remain vigilant about how moral emotions like shame are deployed. A broader perspective on collective responsibility, rather than personal guilt, will be crucial for ensuring that these innovations serve the common good. While shame may never vanish from our moral vocabulary, reimagining its targets and effects offers an opportunity to steer technological progress toward ethically informed and socially responsive practices. By combining collective moral critique with concrete interventions, such as green AI standards, labor protections, and open-source innovation, we can ensure that AI technologies serve as tools for equity rather than engines of exploitation. By reframing shame as a collective force for accountability, we can

reshape AI's trajectory to ensure that technological progress is not an engine of exploitation but a foundation for equity and sustainability. This vision challenges us to confront the systems that perpetuate harm while fostering technological futures rooted in justice, transparency, and solidarity. Only through such a collective commitment can we align innovation with the shared values of equity and responsibility and transform AI from a source of disquiet into a tool for flourishing.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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